# ORLOV'S EQUIVALENCE ON A LOCKED SEXTIC FOURFOLD

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Abstract. Orlov's 2009 theorem is specialized to the sextic $F = x_0^5x_3 + x_3^6 + x_1^5x_4 + x_4^6 + x_2^5x_5 + x_5^6$. Because the Gorenstein parameter vanishes, $D^b(\operatorname{Coh} V(F)) \simeq \mathrm{HMF}^{\mathrm{gr}}(F)$. This is a change of presentation of one category. It is not a Hodge counterexample: field 3 remains empty.

## 1. Disclaimer

This note records an existing theorem on one host. It does not inhabit the type of a rational Hodge counterexample. No function $\mathrm{miss}_\gamma$ is constructed.

## 2. Data

Let $S = \mathbb{C}[x_0, \dots, x_5]$ and $F = x_0^5x_3 + x_3^6 + x_1^5x_4 + x_4^6 + x_2^5x_5 + x_5^6$. Write $X = V(F) \subset \mathbb{P}^5$, assumed smooth, and $R = S/(F)$. Then $n = 5$, $w = 6$, and $a = n + 1 - w = 0$. Hence $K_X \simeq \mathcal{O}_X$.

## 3. The equivalence

**Theorem 1** (Orlov 2009, Thms. 2.5(iii), 2.13(iii), 3.11(iii))**.** $D^b(\operatorname{Coh} X) \simeq D^{\mathrm{gr}}_{\mathrm{Sg}}(R) \simeq \mathrm{HMF}^{\mathrm{gr}}(F)$.

*Proof sketch.* Serre identifies $D^b(\operatorname{Coh} X)$ with $D^b(\operatorname{qgr} R)$. Eisenbud identifies $\mathrm{HMF}^{\mathrm{gr}}(F)$ with the graded singularity category of $R$. Orlov compares the two by window functors $\Phi_i, \Psi_i$, $i \in \mathbb{Z}$. On a sheaf $\mathcal{E}$, $\Psi_i(\mathcal{E})$ is the Eisenbud stabilization of the truncated section module $\tau_{\geq i}\Gamma_*(\mathcal{E}(\bullet))$. When $a = 0$ there is no leftover exceptional collection, so $\Phi_i$ and $\Psi_i$ are inverse equivalences. □

**Remark 1.** *Every window still factorizes the same polynomial $F$.* $\mathrm{ch}(\Psi_i(\mathcal{O}_\Pi)) = [\Pi] \in \mathrm{im}(\mathrm{cl})$.

## 4. What is not proved

Field 3 is the statement that a named primitive class $\gamma \in P^4(X, \mathbb{Q}) \cap H^{2,2}$ lies outside $\mathrm{im}(\mathrm{cl})$. The present equivalence never mentions $\mathrm{CH}^2(X)$. Named classes on this host are algebraic. Clay status remains open.

## References

[1] D. Orlov, Derived categories of coherent sheaves and triangulated categories of singularities, Progr. Math. 270 (2009), 503–531.

[2] D. Eisenbud, Homological algebra on a complete intersection, Trans. Amer. Math. Soc. 260 (1980), 35–64.

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